

MANAGEMENT OF MINOR FRACTURES, SPRAINS AND STRAINS **OF THE FOOT AND ANKLE**

Key Family Physicians:

Dr Ma Yueyun, Dr Tan Juanmin, Dr Amaris Lim

Key Contributors:

Dr Valerie Tan Huali, Dr Zhang Zhi Peng, Dr Sky Koh Wei Chee

Specialist Advisor:

Dr Hong Choon Chiet (Consultant, Orthopaedic Surgery, NUHS)

Acknowledgement for their inputs:

NUHSD: Ms Chua Chie Sing

Adapted from original SFP article
with permission from authors:

Dr Kanwar Sudhir Lather (Emergency Medicine, NUHS)
A/Prof Malcolm Mahadevan (Emergency Medicine, NUHS)

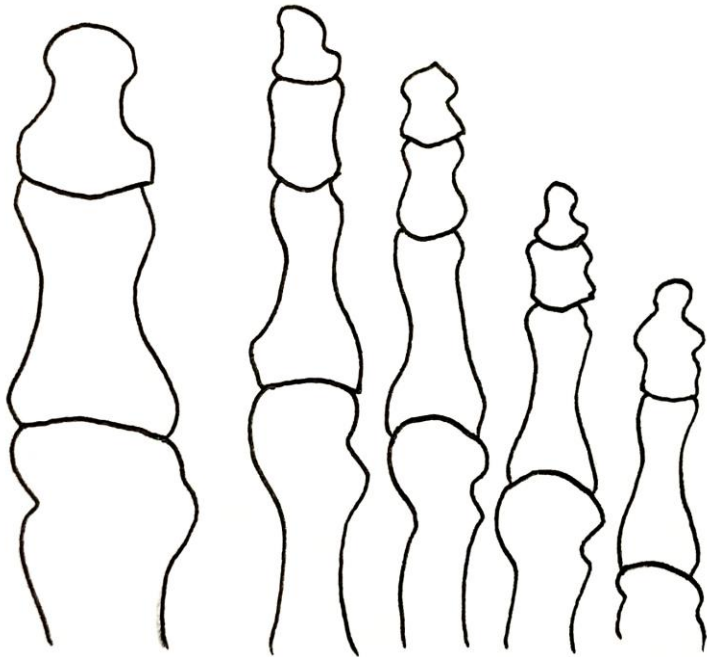
CONTENTS

Introduction	3
Red Flags	3
Toe Fractures.....	4
Anatomy.....	4
Clinical Evaluation	5
Suggested Investigation	5
Management of Toe Fractures	6
Management of Toe Fractures in Polyclinic	8
Metatarsal Fractures.....	9
Acute Fractures	9
Stress Fractures	9
Tarsal Bone Fractures.....	10
Talus, Navicular and Calcaneal Fractures	10
Calcaneal Stress Fractures.....	11
Ankle Sprain	12
Ankle Fractures.....	13
Ottawa Ankle / Foot Rule	13
Follow Ups.....	15
References.....	16

INTRODUCTION

- Strain: A tearing injury to muscle fibres from excessive tension or overuse
- Sprain: A tearing injury to one or more ligament of a joint when joint is forced beyond the limits of its normal plane of motion
- Fracture: A disruption in bone tissue from force exceeding the strength of the bone, repetitive stress or an invasive process
- Principles of management for minor strains, sprains and fractures
 - **P:** Protect – from further damage by using a support or splint (if required)
 - **R:** Rest – and avoid excessive walking or weight bearing on injured side
 - **I:** Ice – for 15 minutes every 2-3 hours (do not apply ice directly to skin)
 - **C:** Compress – with elastic bandage (do not apply when sleeping)
 - **E:** Elevate – and support to reduce swelling and bruising
- When evaluating an injured extremity, always evaluate the joint above and the joint below the site of injury as indirect forces may have affected these sites as well
- Refer **ALL** injuries with **RED FLAGS** to the Emergency Department for further management, including situations where backslab / casts for immobilization are indicated
- **RED FLAGS**
 - Unstable fractures requiring backslab / casts for immobilization, or involving joint lines
 - Fractures involving weight bearing bones or long bones
 - **Possible exception:** small avulsion / chip fractures without significant pain **and not involving a joint**
 - Open fractures or fractures with significant soft tissue injuries (severe nail bed injuries, crush injuries *etc.*)
 - Acute dislocations
 - Any injury with neurovascular compromise
- Fracture lines may not be always visible / obvious in acute fractures
 - If high clinical suspicion (e.g., severe pain, significant swelling, immobility, functional impairment), patient should be managed as for a possible fracture
 - Otherwise, patient can be reviewed and radiographs can be repeated in 1 – 2 weeks after immobilization

TOE FRACTURES



Anatomy

- Great toe
 - 2 Phalanges
 - Crucial role in balance and locomotion (toe-off phase)
- Lesser (2nd to 5th) toes
 - 3 phalanges each, but occasionally may have 2 phalanges (e.g., 4th and 5th toes)
- Each toe has a pair of plantar arteries and nerves, and a pair of dorsal arteries and nerves
- Unusual for the neurovascular structures to be injured as a result of a toe fracture, except in open or severe crush injuries
- Most closed toe fractures can be treated conservatively with excellent outcomes

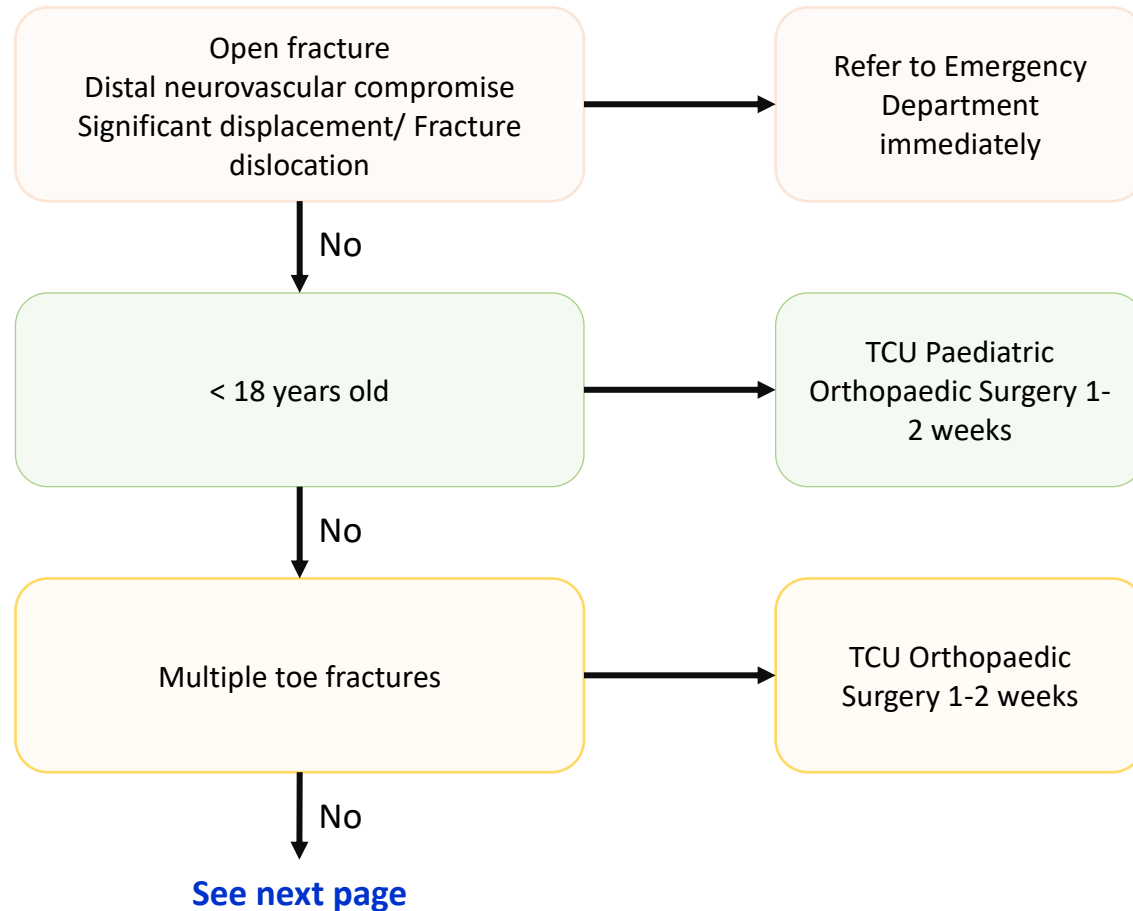
Clinical Evaluation

- Exclude open fractures
- Presence and severity of subungual hematoma. Subungual hematoma is a hallmark of distal phalangeal fractures.
- Clinical toe alignment and rotational deformity
- Neurovascular status

Suggested Investigation

- X-Ray of toe AP and Oblique views

MANAGEMENT OF TOE FRACTURES



Subungual hematoma

- Displaced / fractured nail should be treated as an open fracture
- Intact nail < 48h consider referral to Emergency Department for trephination versus nailbed laceration repair
- If decline referral, counsel on possible loss of nail and nail deformity

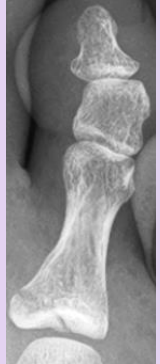
From previous page

No

Great Toe		Lesser Toes
Displaced* Extra-articular	Intraarticular, ≥ 25% joint surface	Displaced* Intra / Extra-articular
		

TCU Orthopaedic
Surgery 1-2 weeks

No

Great Toe		Lesser Toes
Minimally Displaced Extra-articular	Intraarticular, < 25% joint surface	Minimally displaced Intra/ Extra-articular
		

Manage in Polyclinic

Return advice

- Return for review if persistent symptoms after 6-8 weeks

Management of Toe Fractures in Polyclinic

- Analgesia
- Buddy splint of fractured toe to adjacent toe for 1-2 weeks
- Daily elevation and minimise walking on the injured foot in the first 2 weeks
- Avoid sports, jumping or running for 6-8 weeks
- Patient Counselling
 - Most toe fractures heal with no complications within 6-8 weeks
 - Symptoms such as residual stiffness, pain, swelling may last up to 3-6 months post injury
 - Return advice: To return in 6-8 weeks if abovementioned symptoms occur
- Scheduled follow-up for closed, minimally displaced toe fractures are not routinely required
- Reviews in 6-8 weeks can be arranged based on clinician's discretion if any concerns are anticipated
- Offer up to 10-14 days MC, 14 days of light duty (excuse boots) as required

METATARSAL FRACTURES

- Metatarsal fractures can present as acute or stress fractures
- Suggested investigation: XR foot AP and Oblique views
- **Acute Fractures**
 - Most acute metatarsal fractures are treated conservatively with elevation, ice, analgesia and immobilization
 - Non displaced or minimally displaced fractures can be splinted and treated conservatively
 - Significant displacement or angulation will require reduction before immobilization
 - Refer acute metatarsal fractures to the Emergency Department for immobilization (back slab + non-weight bearing)
 - Small avulsion/chip fractures not involving the joint **and** without significant pain can be referred for early Orthopaedic Surgery review in 1-2 weeks
 - Small avulsions/chip fractures **involving the joint** could represent ligamentous injuries such as a Lisfranc injury or metatarsophalangeal joint collateral ligament avulsion and hence should be referred to the Emergency Department for immobilization (back slab + non-weight bearing)
 - **Lisfranc injury can present with seemingly minor X-ray findings (e.g., misalignment of the 2nd MTPJ on X-ray) and should be managed as a potential serious injury with high index of suspicion.**
 - Review with senior doctors if unsure of disposition plan
- **Stress Fractures**
 - Stress fracture of the metatarsal occurs most frequently in the second and third metatarsals
 - Treatment is usually conservative with 6-8 weeks of rest and orthotics to offload metatarsals
 - Calcium and vitamin D supplementation can be started based on clinician's discretion
 - Refer metatarsal stress fractures to Orthopaedic Surgery outpatient clinic in 2-4 weeks



5th metatarsal neck fracture

TARSAL BONE FRACTURES

- **Talus, Navicular and Calcaneal Fractures**

- Mostly occur from high velocity trauma (e.g., fall from height or road traffic accident)
- May be associated with ligamentous injuries or joint dislocations
- Talus and navicular fractures also have an increased risk of avascular necrosis
- Suggested investigation for talus and navicular fractures: XR foot AP and Oblique views + XR ankle joint AP and Lateral views
- Suggested investigation for calcaneal fractures: XR calcaneum Axial and Lateral views
- Refer these injuries to the Emergency Department in the acute setting

Navicular fracture



- **Calcaneal Stress Fractures**

- Usually from repetitive stress over the heel
- Mild symptoms can be managed with activity restriction and heel inserts
- Significant symptoms (e.g., pain and swelling with walking): non-weight bearing with crutches until symptoms subside
- **Suggested investigation: XR calcaneum Axial and Lateral views**
- Calcium and vitamin D supplementation can be started based on clinician's discretion
- Refer stress fractures to Orthopaedic Surgery outpatient clinic in 2-4 weeks



Calcaneal Fracture



Calcaneal Stress Fracture

Published with permission from LearningRadiology.com

LearningRadiology.com (C)
All Rights Reserved

ANKLE SPRAIN

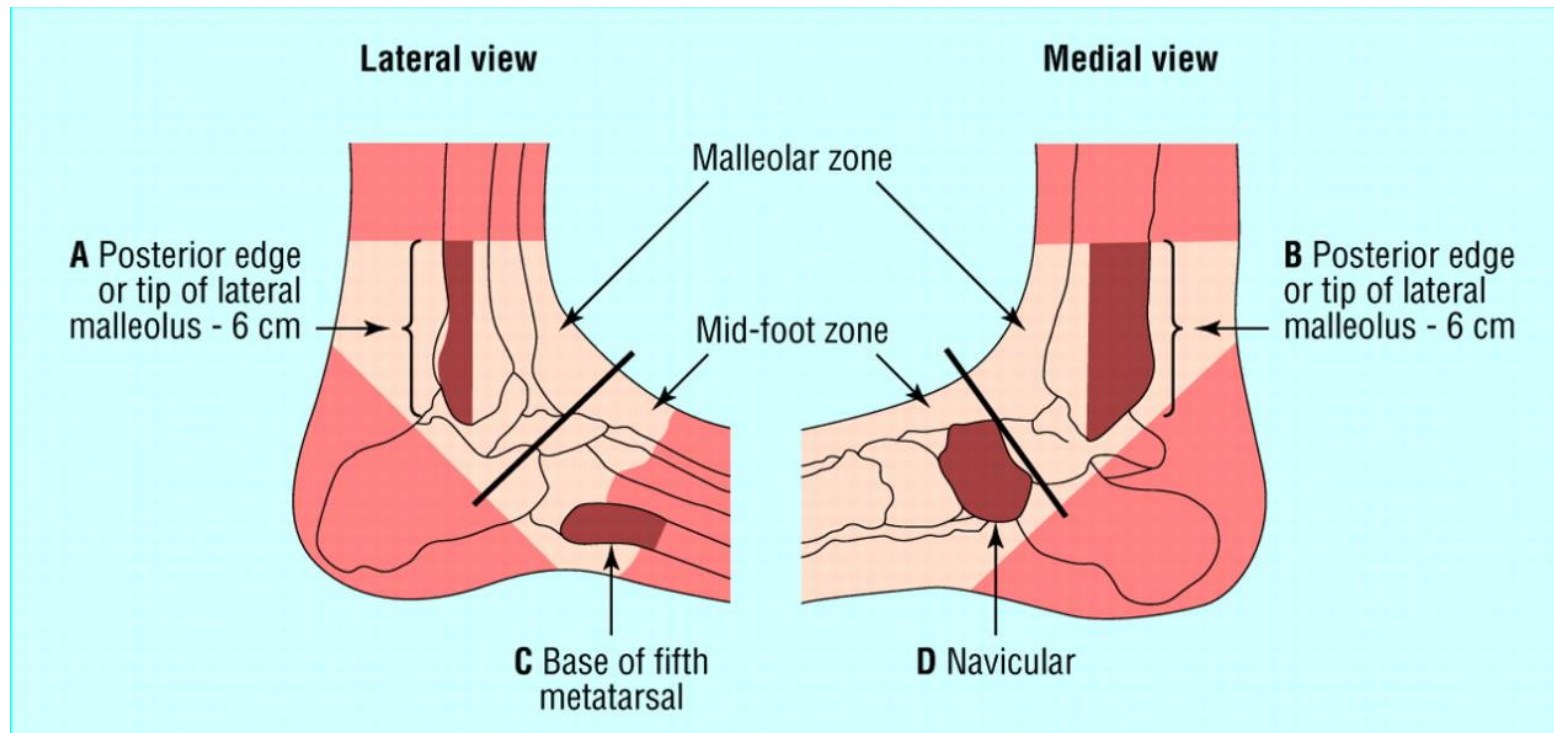
- Ankle sprains are among one of the commonest sports injuries
- Exclude ankle fracture with targeted physical examination and appropriate imaging (refer to [Ankle Fracture Section](#))
- Initial management includes **protection**, rest, ice, **compression and elevation**
 - **Patients can be sent to nursing team for bandaging of the sprained ankle as part of management**
- **Patients should be referred to physiotherapy for rehabilitation**
- Majority of patients will recover well with rest and functional rehabilitation such as proprioceptive training, peroneal tendon strengthening and stretching ± ankle bracing. Patients with residual symptoms of ankle instability, recurrent sprains, pain and swelling upon re-attendance should be referred to Orthopaedic Surgery in 4-6 weeks
- If the swelling and bruising is out of proportion to the nature of trauma, an occult fracture should be suspected
 - The patient may benefit from temporary immobilization with and non-weight bearing for 1-2 weeks
 - Refer to the Emergency Department for further assessment and intervention if necessary

ANKLE FRACTURES

- Ankle fractures are among one of the commonest orthopaedic injuries
- **Suggested investigation: XR ankle joint AP and Lateral views**
- The Ottawa Rule can be applied to determine indication for radiograph in ankle and foot injuries

Ottawa Ankle / Foot Rule

- Radiographs of **ANKLE** is only required if:
 - Any pain in the malleolar region
 - **PLUS**
 - Bony tenderness at the distal posterior edge of fibula (6 cm) or tip of lateral malleolus
 - **OR**
 - Bony tenderness at the distal posterior edge of tibia (6 cm) or tip of medial malleolus
 - **OR**
 - Inability to bear weight (limping counts as bearing weight) both immediately and in consult room for 4 steps
- Radiographs of **FOOT** is only required if:
 - Any pain in the midfoot region
 - **PLUS**
 - Bony tenderness at the base of the fifth metatarsal
 - **OR**
 - Bony tenderness at the navicular
 - **OR**
 - Inability to bear weight (limping counts as bearing weight) both immediately and in consult room for 4 steps



- Acute ankle fractures should be referred to the Emergency Department for immobilization (back slab + non-weight bearing)
 - Small avulsion / chip fractures without significant pain may not require Emergency Department referral but early Orthopaedic Surgery review in 1-2 weeks
 - Can consider other forms of immobilization (e.g., ankle brace)
 - Review with senior doctors if unsure of disposition plan

FOLLOW UPS

- Scheduled follow-up for closed, minimally displaced toe fractures are not routinely required
- In general, non-toe fractures of the lower limbs are referred to Emergency for back slab or acute management, or early Orthopaedics review after stabilisation (if achievable) in the primary care setting
- Depending on the severity and region of injury, bony and ligamentous injuries can take up to 6-9 months to heal. Symptoms usually improves progressively after the initial 6 to 8 weeks
- Patient who initially presented with negative radiographic findings but with persistent symptoms such as pain or swelling should have a repeat assessment
 - Repeat radiographs to ensure no fracture was missed
 - Ensure adequate pain control and compliance to rest, elevation and weight bearing status
- Consider specialist referral in the presence of:
 - Worsening or persistent pain, swelling or loss of function / disability due to the fracture
 - Malunion, non-union, or delayed union of the fracture site
 - Injuries where injury claim / compensation or legal input may be required
 - Any outstanding physician or patient concern

REFERENCES

1. Kanwar Sudhir Lather, Malcolm Mahadevan. **Minor fracture, sprain and strain**. The Singapore Family Physician Vol 41(3) Jul-Sept 2015
2. Diagnosis and Management of Common Foot Fractures. Am Fam Physician. 2016 Feb 1;93(3):183-191
3. Vliet-Koppert, S. T. V., Cakir, H., Lieshout, E. M. V., Vries, M. R. D., Elst, M. V. D., & Schepers, T. (2011). Demographics and Functional Outcome of Toe Fractures. *The Journal of Foot and Ankle Surgery*, 50(3), 307–310. doi: 10.1053/j.jfas.2011.02.003
4. Mittlmeier, T., & Haar, P. (2004). Sesamoid and toe fractures. *Injury*, 35(2), 87–97. doi: 10.1016/j.injury.2004.07.015
5. Adapted from: Hatch, R. L., Hacking, S. (2003). Evaluation and Management of Toe Fractures. Am Fam Physician, 68(12):2413-2418.
6. Garcia, M. R., Nicholson, A. D., Nduaguba, A. M., Sanders, J. O., Liu, R. W., & Cooperman, D. R. (2018). Ossification of the phalanges of the foot and its relationship to peak height velocity and the calcaneal system. *Journal of Childrens Orthopaedics*, 12(1), 84–90. doi: 10.1302/1863-2548.12.170164
7. York, P. J., Wydra, F. B., & Hunt, K. J. (2017). Injuries to the great toe. *Current Reviews in Musculoskeletal Medicine*, 10(1), 104–112. doi: 10.1007/s12178-017-9390-y
8. Hong CC, C J Pearce, M S Ballal, J D F Calder. Management of Sports Injuries of the Foot and Ankle: An Update Bone Joint J. 2016 Oct;98-B(10):1299-1311. doi: 10.1302/0301-620X.98B10.37896.
9. Hong CC, Tan KJ. Concepts of ankle instability: A review. OA Sports Medicine 2014 Feb 22;2(1):3.